



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510 Fax: +66-2721-3515

SAFETY DATA SHEET

Company Identification

Thep Watana Co., Ltd. Address: 382 Bangpoo Industrial Estate.

Sukhumvit Road, Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66(0)-2721-3510; Fax: +66(0)-2721-3515

1. CHEMICAL IDENTIFICATION

Product Identification

Trade name: Re-ong

Product name: haloxyfop-P-methyl 10.8% W/V EC

Product use: Herbicide

Relevant identified uses of the substance/mixture and uses advised against:

Used as a systemic, post-emergent, selective herbicide for the control of many annual and perennial grass weeds in a number of crops.

2. HAZARD IDENTIFICATION

Potential Health Effects

Swallowed: The acute oral toxicity is low and small amount ingested incidentally during the handling of the concentrate and use of prepared spray mix is not likely to be harmful.

Eye: Solvent vapor may irritate the eyes. Concentrate in eye may cause moderate to severe irritation and possible corneal injury. The prepared spray in eye may cause moderate to severe irritation and possible corneal injury. The prepared spray mix may be slightly irritating to the eye.

Skin: The acute dermal toxicity is low and a single prolonged skin exposure is not likely to result in absorption of harmful amounts.

Inhaled: The acute inhalation toxicity is low and absorption of harmful amounts by inhalation is unlikely.

3. COMPOSITION/INFORMATION ON INGREDIENT

Chemical name: Methyl(*R*)-2-[4-(3-chloro-5-trifluoromethyl-2-pyridyloxy)phenoxy]propanoate
[IUPAC]

ISO Common name: Haloxyfop-P



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CAS Number: 72619-32-0

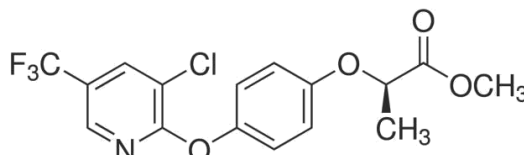
Ingredients and composition: Active ingredient – haloxyfop-P-methyl 10.8% W/V
Other ingredient – to 1 L

(Ingredients not precisely identified are proprietary or non-hazardous)

Formula: $C_{16}H_{13}ClF_3NO_4$

Molecular weight: 375.7 g/mol

Structural formula:



4. FIRST AID MEASURES

Description of first aid measures

In case of swallowed:

If swallowed, give water to dilute. DO NOT induce vomiting unless medical attention is delayed more than one hour. Take to a doctor or hospital immediately.

In case of eyes contact:

Immediately flush with cold water. Continue for at least 15 minutes. Contact a doctor.

In case of skin contact:

Remove contaminated clothing and wash skin thoroughly with soap and water. DO NOT scrub the skin.

In case of Inhaled contact:

If ill-effects occur move to fresh air. Contact a doctor.

Note to Physician:

Gallant NF contains petroleum-like solvents and the benefits of inducing vomiting must be weighed against possibility of chemical pneumonitis. If lavage is performed, endotracheal and/or esophageal control is advisable.

5. FIRE FIGHTING MEASURES

Fire & Explosion Hazard:

Flammable solvent in formulations. Combination of the vapor from the solvents with air may form an explosive mixture. Rapid decomposition of the active ingredient occurs at temperatures greater than 300 °C resulting in the formation of hazardous decomposition products. Drum explosion is likely if confined at high temperatures. Intact containers exposed to foam blanket to contain product. Avoid using AFFF. Contain runoff. Breathing apparatus is required.



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6. ACCIDENTAL RELEASE MEASURES (SPILLAGE)

Small spill:

Wear appropriate protective clothing whilst cleaning up small spills (see personal Protection).

Apply absorbent material such as earth, sand or clay granules to the spill. Sweep up material for disposal when absorption is complete.

Large spill:

Wear protective clothing and a respirator suitable for organic vapors. Clear the area of all unprotected personnel. As a standard precautionary measure, shut off all sources of ignition. Place leaking containers into salvage drums. Apply absorbent material such as earth, sand or clay granules to spill area. Form a barricade around the spill and in front of drains or waterways in spill vicinity, using earth or other available material. Prevent entry of material into drains or water ways.

7. HANDLING AND STORAGE

Storage and Transport:

Not classified as a dangerous good. Store in tightly closed original containers in a locked, well ventilated place, out of direct sunlight. Avoid contact with food, clothing, drugs or other household goods during transport.

8. EXPOSURE CONTROL/PERSONAL PROTECTION

Engineering Control

End user: Not special controls required for normal conditions of use. Prepare spray in well ventilate area.

Manufacturing: Atmospheric levels should be maintained below the recommended exposure guide-lines. Local exhaust ventilation may be necessary for some operations.

Personal Protection

End User:

When preparing the spray, wear overalls buttoned to the neck and wrist, a washable hat and elbow length PVC gloves, face-shield or goggles. When using the prepared spray, wear overalls buttoned to the neck and wrist, a washable hat and face-shield or goggles. Wash hands after use. After each day's use wash gloves, face-shield or goggles and contaminated clothing.

Manufacturing:

Wear overalls buttoned to the neck and wrist, a washable hat and elbow length PVC gloves, face-shield or goggles.



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9. PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

These physical data are typical values based on material tested but may vary from sample to sample. Typical values should not be construed as a guaranteed analysis of any specific lot or as specifications for the product.

Physical stage:	Liquid
Color:	Yellow brown to red brown
Odor:	Light characteristic odor
Melting point:	Not available
Specific gravity:	1.010 ± 0.005 g/cm ³
Vapor pressure:	Not available
Solubility in water:	Emulsifiable
Partition coefficient:	log Pow = 4.00 (technical)

10. STABILITY AND REACTIVITY

Stability

Stable under normal storage condition.

Hazardous decomposition products

Can decompose at high temperatures forming toxic gases.

Conditions to avoid

Extreme temperature.

Hazardous polymerization

Not pertinent.

11. TOXICOLOGICAL INFORMATION

Information on Acute toxicity for end product, Haloxyfop-P-methyl:

LD50 (Oral, rat):	> 2,000 mg/kg body weight
LD50 (Dermal, rat):	> 2,000 mg/kg body weight
LC50 (Inhalation, rat, 4h):	> 4.74 mg/L air
Skin irritation (rabbit):	Slightly irritating to rabbit skin.
Eye irritation (rabbit):	Moderately irritating to rabbit eye.
Skin sensitization:	Not classified as sensitizer



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12. ECOLOGICAL INFORMATION

Toxicity

Adopt good working practices, so that the product is not released into the environment.

Data referred to the mixture, Haloxyfop-P-methyl:

Fish – Acute toxicity:

Rainbow trout,

LC50 (96 h): 0.7 mg/L

Invertebrates – Acute toxicity:

Algae,

EC50 (96 h): > 70 mg/L

Birds – Toxicity:

Bobwhite quail,

LD50: 414 mg/kg

Arthropod – Toxicity:

Bee,

LD50 (oral/contact, 48 h): > 100 ug/bee

Soil organism, invertebrates – Toxicity:

Earthworm,

LC50 (14 days): 1,343 mg/kg

13. DISPOSAL CONSIDERATIONS

Disposal:

Contaminated material must be disposed of in accordance with national, regional and district requirements.

Small Quantities and Container:

Bury contaminated material at an approved landfill. Puncture top, bottom and sides of containers and bury below 500 mm in a disposal pit specifically marked and set up for this purpose, clear of waterways, vegetation and roots.

14. TRANSPORT INFORMATION

The data provided in this section is for information only. Please apply the appropriate regulations to properly classify your shipment for transportation.

UN No: 3082

Dangerous Goods Class: 9



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Packaging Group:	III
Marine pollutant:	No
Proper shipping name:	Environmentally Hazardous Substance Liquid N.O.S.

15. REGULATORY INFORMATION

Designated name:

Haloxypop-P-methyl 10.8% W/V EC

Categories of danger

Harmful. Dangerous for the environment.

EC hazard

Xn - harmful

16. OTHER INFORMATION

This MSDS summarizes our best knowledge of the health and safety hazard information of the product and how to safely handle and use the product in the workplace. Each user should read this SDS and consider the information in the context of how the product will be handled and used in the workplace including in conjunction with other products.